

SafeSpeak: Real-Time Cyberbullying Detection Before Social Media Posting

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ABSTRACT

Cyberbullying remains a pressing challenge on social media platforms, causing severe psychological distress to users. This research presents an applied solution for real-time cyberbullying detection, aiming to prevent the posting of harmful content before it appears online. A comprehensive dataset of over 57,000 annotated social media comments, spanning Twitter, YouTube, and various public sources, was aggregated and preprocessed through state-of-the-art text cleaning and normalization techniques. The study compares traditional machine learning models—Logistic Regression, Multinomial Naive Bayes, and Linear SVC (with TF-IDF text representation)—against a fine-tuned BERT transformer for binary cyberbullying classification. Experimental results demonstrate that the BERT model achieves the highest accuracy (0.84), outperforming classical baselines in all metrics, confirming its superiority for nuanced linguistic detection tasks. The system is deployed in a prototype GUI ("Safe Keyboard") that provides instant user warnings, enabling practical pre-posting intervention. This paper empirically validates the effectiveness of both approaches, emphasizing real-world applicability for safer digital communication.

KEYWORDS: *Cyberbullying Detection, Real-time Moderation, Social Media, BERT, Machine Learning, Natural Language Processing, Text classification, Pre-posting Intervention, Content Moderation, Safe Keyboard*

1. INTRODUCTION

Cyberbullying is a widespread problem on social media and digital communication platforms, impacting individuals' psychological health and contributing to hostile online environments. Although platforms employ reactive strategies to moderate harmful content, these methods typically detect and remove abusive messages only after they have been posted, allowing initial damage to occur.

This research addresses the need for proactive, real-time interventions by developing a system that evaluates user-generated text (comments, tweets, direct messages) for cyberbullying intent before it is published. By leveraging advanced machine learning and deep learning techniques, the aim is to prevent potentially harmful content from ever reaching its intended target.

The project aggregates over 57,000 social media messages from diverse sources, applying rigorous preprocessing to construct a high-quality annotated corpus. It compares traditional models such as Logistic Regression, Multinomial Naive Bayes, and Linear SVC (using TF-IDF text features) against a fine-tuned BERT transformer for nuanced text classification.

A novel contribution is the integration of these models into a user-friendly, real-time "Safe Keyboard" interface that provides immediate feedback to users, blocking cyberbullying attempts before posting. This work demonstrates the feasibility, accuracy, and impact of pre-posting moderation and sets the stage for future deployment of such systems across public social platforms.

2. LITERATURE REVIEW

Cyberbullying detection has become a critical research area in natural language processing and social computing, driven by the increasing prevalence of harmful online behavior. Recent studies have approached this challenge through various machine learning and deep learning methodologies.

2.1. Traditional Machine Learning Approaches

Early research in cyberbullying detection primarily employed classical machine learning algorithms coupled with text representation techniques such as term frequency-inverse document frequency (TF-IDF) and bag-of-words models. Several studies demonstrated that

ensemble methods, particularly voting classifiers combining Logistic Regression, Support Vector Machines (SVM), and Naive Bayes, achieve accuracies up to 96.5%. Singh et al. reported that XGBoost outperformed traditional algorithms, achieving 97% accuracy on Twitter data using TF-IDF representations.

Although effective, these approaches often fail to capture deeper semantic nuances and contextual dependencies in textual content, limiting their performance in dynamically evolving social media contexts.

2.2. Deep Learning and Transformer-Based Models

The introduction of transformer architectures, particularly BERT (Bidirectional Encoder Representations from Transformers), has significantly enhanced the capability of automated cyberbullying detection. Recent studies report BERT-based models attaining between 90–94% accuracy across various social media datasets. Muralidhar (2024) demonstrated that BERT classifiers achieved 94% accuracy on labeled cyberbullying datasets, underscoring the model’s strength in understanding contextual linguistic patterns.

Hybrid methods integrating BERT with classical machine learning have also shown promising outcomes. A 2024 study developed an ensemble model combining BERT and SVM with grid search optimization, achieving 90% accuracy for multiclass cyberbullying classification. Similarly, Kumar et al. (2025) introduced a feature fusion approach incorporating TF-IDF, Word2Vec, and BERT embeddings, reporting a 15% accuracy improvement over single-feature baselines and demonstrating enhanced robustness in identifying nuanced linguistic patterns such as sarcasm and coded slurs.

2.3. Pre-Posting Interventions and Real-World Deployments

Although detection accuracy has improved considerably, relatively few studies have explored real-time pre-posting intervention—a critical phase between message creation and public dissemination. Existing content moderation systems largely focus on post-publication filtering or user reporting. One notable exception is the “ReThink” intervention tool, which proactively alerts users about potentially harmful messages prior to posting, yielding measurable reductions in offensive content transmission.

Recent work stresses the necessity for proactive moderation systems that function at the user interface level. Proposed frameworks include automated blocking tools embedded within social media applications or as standalone utilities to prevent cyberbullying attempts before reaching victims. Nonetheless, several deployment challenges remain unresolved, including latency limitations affecting real-time inference, false positive risks that could impede free expression, and the design of intuitive interfaces that promote user acceptance without hindering communication flow.

2.4. Research Gaps and Positioning of This Work

Despite substantial progress in algorithmic development, few studies provide systematic comparisons between classical machine learning models (TF-IDF combined with SVM, Logistic Regression, or Naive Bayes) and transformer-based models (BERT) under identical datasets and preprocessing pipelines. Furthermore, the majority of research remains confined to experimental or prototype stages, with minimal emphasis on practical deployment mechanisms.

The present study addresses these deficiencies by conducting a rigorous comparative analysis of traditional and deep learning methods using a unified dataset comprising over 57,000 social media posts from multiple platforms. Additionally, the design and implementation of a “Safe Keyboard” graphical user interface (GUI) prototype demonstrate the practical feasibility of real-time pre-posting intervention, effectively bridging the gap between theoretical research and applied social media safety mechanisms.

3. PROBLEM STATEMENT

The surge of user-generated content on social media platforms has exacerbated incidents of cyberbullying, inflicting serious emotional and psychological harm on victims before any mitigation measures take effect. Existing moderation systems are predominantly reactive—they identify and remove abusive content only after it has been posted—thereby failing to prevent the initial exposure and impact on targets.

This research seeks to develop a proactive, real-time cyberbullying detection system that intervenes at the point of content creation. Specifically, the system must:

- Accurately identify potentially harmful text messages, comments, tweets, or direct messages before publication.
- Operate with low latency to provide immediate feedback without disrupting user experience.
- Support diverse language patterns by leveraging both TF-IDF-based classical models and context-aware transformer architectures.
- Integrate seamlessly into user interfaces (e.g., keyboard extensions or chat boxes) to block or warn users when cyberbullying content is detected.

By addressing these requirements, the project will shift the moderation paradigm from reactive post-hoc filtering to a preventive intervention that curtails cyberbullying at its inception, thereby fostering safer online communication environments.

4. System Overview

This section outlines the architecture and workflow of the proposed real-time cyberbullying detection system.

4.1. High-Level Architecture

The proposed system employs a modular design architecture composed of three primary components: the input interface, the processing engine, and the feedback and action module. The input interface incorporates a “Safe Keyboard” implemented using Tkinter, which captures user input securely before submission. This controlled input acquisition ensures that all text data is obtained in a safe and consistent manner.

The processing engine serves as the core analytical framework, consisting of three sequential modules: preprocessing, feature extraction, and classification. The preprocessing module prepares the text data by converting emojis into their text descriptions using the *demoji* library, transforming all characters to lowercase, and removing extraneous elements such as URLs, mentions, and hashtags. Additionally, punctuation stripping, whitespace normalization, tokenization, stopword elimination, and lemmatization are applied to simplify and standardize the text, particularly for classical machine learning models.

Feature extraction is conducted differently based on the classification approach. For traditional machine learning models, Term Frequency-Inverse Document Frequency (TF-IDF) vectorization is utilized to statistically represent the text features. In contrast, for the transformer-based model, a BERT tokenizer with a maximum sequence length of 64 is employed to capture semantic and contextual information essential for deep learning classifiers.

The classification module integrates both conventional machine learning algorithms and a transformer architecture. Specifically, Linear Support Vector Classifier (SVC), Logistic Regression, and Multinomial Naive Bayes classifiers are employed as classical methods, while a fine-tuned *BERT-base-uncased* model constitutes the advanced transformer-based approach. This combination enhances the robustness and accuracy of cyberbullying detection.

Finally, the feedback and action module produces a binary classification output, categorizing user input as either cyberbullying or safe content. The graphical user interface provides instantaneous visual feedback: a red warning overlay signals the detection of harmful content, whereas a green confirmation indicates safe text. Users are given the option to either block

the submission of flagged messages or to rephrase their input in accordance with system prompts.

4.2. Workflow Diagrams



Fig.1

4.3. Component Descriptions

The system integrates a Tkinter-based graphical user interface, referred to as the “Safe Keyboard,” which embeds seamlessly into the user environment. This interface intercepts text inputs in real time and displays classification results inline, allowing immediate feedback. Separate implementations of the keyboard exist to support both traditional machine learning (ML) models and a fine-tuned BERT transformer model, ensuring compatibility with the underlying classification approach.

The preprocessing pipeline standardizes input text across models by encapsulating all necessary data cleaning steps. For traditional ML models, additional preprocessing is performed, including stopword removal and lemmatization, to enhance feature quality. In contrast, the BERT model requires minimal preprocessing due to its contextual embedding capabilities, preserving richer semantic content.

To evaluate system performance, a comparative study was conducted using lightweight ML classifiers—namely Linear Support Vector Classifier (SVC), Logistic Regression, and Naive Bayes with TF-IDF features—against a fine-tuned BERT transformer model. This empirical comparison focused on metrics such as classification accuracy, inference speed, and feasibility of deployment in real-time environments.

For real-time inference, models are served locally on a CPU, enabling instant classification and demonstrating the practical deployment potential for pre-posting interventions that can mitigate harmful content proactively. This design not only facilitates rigorous empirical evaluation of classical versus deep learning methods but also validates the viability of deploying functional GUI prototypes for real-world applications.

5. DATA

5.1. Data Sources

Five labeled datasets aggregated into a single corpus of 57,466 samples:

- Twitter Racism Dataset (13,471 samples)
- Twitter Sexism Dataset (14,881 samples)
- Generic Twitter Dataset (16,851 samples)
- YouTube Comments Dataset (3,464 samples)
- Cyberbullying Kaggle Dataset obtained from “cyberbullying-dataset” on Kaggle (8,799 samples)

5.2. Data Collection

Each dataset was downloaded in CSV format, parsed to retain only text and binary labels (cyberbullying vs. non-cyberbullying), and merged into a master file.

5.3. Preprocessing and Cleaning

- Emoji Handling: Converted to text descriptions using *demoji*
- Case Normalization: All text lowercased
- Noise Removal: URLs, mentions (@user), and hashtags (#tag) stripped via regex
- Punctuation & Whitespace: Removed punctuation, normalized whitespace
- Tokenization: Word-level tokenization (`word_tokenize`) for classical models
- Stopword Removal & Lemmatization: Applied only for TF-IDF-based models using NLTK’s English stopwords list and WordNetLemmatizer

5.4. Dataset Characteristics

- Total Samples: 57,466
- Label Distribution: Balanced across two classes (approx. 50% cyberbullying, 50% non-cyberbullying)
- Platform Diversity: Mix of Twitter and YouTube user-generated content

This multi-source aggregation ensures robust model training and evaluation across varying styles, topics, and contexts.

6. METHODOLOGY

6.1. Preprocessing Pipeline

The preprocessing pipeline begins by converting all emojis in the text into descriptive text representations using the *demoji* library. Subsequently, the entire text is converted to lowercase to standardize input casing. To remove noise, URLs (matching patterns such as *http(s)://...* and *www...*), mentions (@username), and hashtags (#hashtag) are stripped using regular expressions. Punctuation characters are then removed utilizing Python’s

string.punctuation module. The pipeline normalizes whitespace by collapsing multiple consecutive spaces into a single space and trimming leading or trailing spaces.

For classical machine learning models, tokenization is performed using NLTK's *word_tokenize*, followed by the removal of English stopwords from NLTK's stopword list. Tokens are then lemmatized using the *WordNetLemmatizer*. For transformer-based BERT models, only the initial five preprocessing steps are executed before tokenization with the BERT tokenizer.

6.2. Feature Extraction

Feature extraction differs between classical and transformer-based models. Classical models convert the preprocessed text into term frequency–inverse document frequency (TF-IDF) vectors using scikit-learn's *TfidfVectorizer* with default parameters. In contrast, the transformer model encodes text using the *BertTokenizer* (*bert-base-uncased*) with a maximum sequence length of 64 tokens, applying necessary padding or truncation.

6.3. Model Descriptions

Three classical machine learning classifiers are employed: Logistic Regression, Multinomial Naive Bayes, and Linear Support Vector Classifier (SVC). Logistic Regression with L2 regularization is implemented via scikit-learn's *LogisticRegression*. Multinomial Naive Bayes, a probabilistic model based on term frequencies, is realized through scikit-learn's *MultinomialNB*. The Linear SVC with a linear kernel utilizes scikit-learn's *LinearSVC*.

The transformer-based approach involves fine-tuning the *BertForSequenceClassification* model (comprising approximately 110 million parameters). The model employs a binary classification head implemented using PyTorch and the Hugging Face *transformers* library.

6.4. Training Protocols

Datasets are divided into training and testing subsets with an 80:20 ratio. For fine-tuning BERT, training is conducted for four epochs using the AdamW optimizer with default weight decay parameters. Learning rate scheduling is handled by *get_linear_schedule_with_warmup*, configured with zero warm-up steps and the total number of training steps equal to the product of the training dataloader length and number of epochs.

Training uses a batch size of 32 and a learning rate of 2×10^{-5} . A fixed random seed of 42 ensures reproducibility across Python's *random* module, NumPy, and PyTorch (CPU and CUDA). Gradient clipping with a norm threshold of 1.0 is applied to stabilize training. The training environment is a Google Colab GPU, with each run taking approximately 18 minutes. Metrics recorded per epoch include training loss, validation loss, training time, and validation time.

For the classical machine learning models, the same train/test split is applied. Logistic Regression is configured with default parameters (*solver='liblinear'*, *C=1.0*). Linear SVC is

trained with default regularization ($C=1.0$, $tol=1e-4$), while Multinomial Naive Bayes employs a default smoothing parameter ($\alpha = 1.0$). Feature extraction for these models is performed using TF-IDF vectors produced via scikit-learn’s default configuration.

7. RESULTS AND DISCUSSION

7.1. Quantitative Results

Model	Accuracy	Precision (Cyberbullying)	Recall (Cyberbullying)	F1-Score (Cyberbullying)	Avg. Inference Latency
Logistic Regression (TF-IDF)	0.78	0.75	0.79	0.77	~30 ms
Multinomi al Naive Bayes	0.72	0.70	0.74	0.72	~25 ms
Linear SVC (TF-IDF)	0.80	0.78	0.81	0.79	~35 ms
BERT (Fine-tune d)	0.84	0.82	0.85	0.83	~200 ms

7.2. Comparative Analysis

The comparative analysis between classical machine learning models and the BERT transformer model reveals several key findings. The BERT model outperforms all classical baselines in terms of accuracy and F1-score by approximately 4 to 12 percent, demonstrating its enhanced ability to capture contextual cues and nuanced forms of abusive language. This highlights BERT's effectiveness in complex natural language understanding tasks critical for cyberbullying detection.

However, this increase in performance comes at the cost of higher latency. Specifically, the BERT model incurs an average inference time of around 200 milliseconds, in contrast to sub-50 millisecond response times observed with classical machine learning methods. This distinction suggests that traditional models may be more appropriate for deployment in environments with stringent real-time latency constraints.

In terms of precision and recall balance, classical models generally provide more evenly matched results, while BERT achieves superior recall. The higher recall rate of BERT is particularly important for minimizing false negatives—instances where cyberbullying content might be missed. These findings emphasize the need to carefully consider the trade-offs between accuracy, response time, and detection priorities when selecting a model for practical applications.

7.3. Error Analysis

The error analysis of the cyberbullying detection system reveals several critical challenges that affect classification accuracy. Firstly, the BERT model occasionally produces false positives by flagging benign messages that contain slang or strong language used out of context as harmful. This limitation indicates that even advanced transformer models can struggle with interpreting informal or colloquial expressions accurately.

In contrast, classical machine learning models are more prone to false negatives, failing to detect subtle bullying instances that involve implicit threats or sarcasm. These nuances are difficult to capture using simpler feature extraction methods, which restricts the sensitivity of traditional approaches.

Additionally, labeling inconsistencies present in several source datasets contribute to noise during model training and evaluation. Mislabelled examples—where non-abusive comments were tagged as cyberbullying and vice versa—introduce ambiguity and reduce overall model reliability. This issue underscores the importance of rigorous data curation and verification processes to enhance the effectiveness of cyberbullying detection systems.

8.4. Discussion

The empirical comparison confirms that transformer-based models significantly enhance detection performance in cyberbullying applications. These models demonstrate superior

accuracy in capturing nuanced and contextual language features compared to classical methods. However, the increased computational cost associated with transformer models requires careful deployment strategies to balance accuracy and system responsiveness.

A viable approach to managing this trade-off is the implementation of hybrid strategies. For example, under conditions of high computational load, the system can fallback to classical machine learning models, which provide faster inference times, thereby maintaining real-time responsiveness without severely compromising detection quality.

The prototype “Safe Keyboard” validates the practical feasibility of this approach. It offers immediate user feedback on potentially harmful content without noticeable typing disruption. This responsive user experience indicates that the system is ready for further stages of evaluation, including user studies and integration into larger platforms.

8. APPLICATIONS

The system is packaged as a desktop prototype called Safe Keyboard, implemented with Python’s Tkinter library. Upon launching, the GUI presents a text entry widget styled as a chat input box:

8.1. Text Capture and Model Invocation

All keystrokes are monitored in real time by the system. Upon completion of each sentence, triggered by the entry of a space or punctuation, the current input text is processed through a preprocessing pipeline. This cleaned input is then passed to either a serialized Joblib classifier for classical machine learning models or a PyTorch checkpoint for the fine-tuned BERT model.

8.2. Prediction and User Feedback

If the model detects potential cyberbullying content, the input box border changes to red, and a warning label appears below, prompting the user to revise their text. If the input is considered safe, the border remains green, and the “Send” button is enabled to allow submission.

8.3. Intervention and User Experience

To prevent the submission of harmful content, the system blocks message sending until the user edits any flagged text. This approach seamlessly embeds cyberbullying detection into the user’s typing workflow, offering pre-posting intervention with minimal latency and an uninterrupted user experience.

9. LIMITATIONS

The system faces several important limitations that impact its current performance and deployment scope. First, label noise due to mislabelled instances in the source datasets has introduced training noise, which adversely affects the overall accuracy and robustness of the models. Such inconsistencies in ground truth reduce the reliability of both classical and transformer-based classifiers.

Secondly, latency constraints present a challenge, particularly with the BERT model's inference time of approximately 200 milliseconds per message. This delay may be excessive for environments requiring high-throughput processing or mobile device deployment, where responsiveness is critical.

Additionally, the prototype "Safe Keyboard," implemented using Python's Tkinter library, is currently a desktop-only proof of concept. Its integration into other platforms, such as browser extensions or mobile applications, has yet to be tested or developed, limiting its practical applicability.

Lastly, language coverage is an inherent constraint. The pretrained BERT model and associated NLTK preprocessing pipeline are designed primarily for English text. As a result, the system may struggle to accurately detect cyberbullying in code-mixed or non-English content, which is common in many multilingual user bases.

10. FUTURE WORK

The system's future enhancements focus on improving data quality, model efficiency, language coverage, user interaction, and deployment scalability. Data quality improvement is paramount; efforts should include curating and relabeling noisy dataset examples while employing active learning techniques to iteratively refine dataset accuracy and model robustness. This approach can mitigate training noise and improve classifier reliability.

Model optimization is another critical area. Research into lightweight transformer variants such as DistilBERT and MobileBERT or techniques like knowledge distillation can help reduce inference latency without significant accuracy degradation, making the system more suitable for real-time, resource-constrained environments.

To address limitations in language support, fine-tuning multilingual transformer models like mBERT or XLM-R is recommended. These models can facilitate detection of cyberbullying in code-mixed and non-English text, thus broadening the system's applicability to diverse linguistic populations.

Conducting user studies is essential to evaluate the practical effectiveness and usability of the pre-posting warning system. Such human factors assessments within real-world social media contexts will provide insights into user acceptance and behavioral impact.

Finally, scaling the system requires platform integration beyond the desktop proof of concept. Developing browser extensions and mobile keyboard plugins will enable wide deployment across popular social media platforms, ensuring broader reach and real-world utility.

11. CONCLUSION

This work presents SafeSpeak, a real-time cyberbullying detection framework that intervenes at the point of content creation, comparing traditional TF-IDF-based classifiers against a fine-tuned BERT transformer. Experimental results on a diverse corpus of 57,466 social media messages demonstrate that BERT achieves the highest detection performance (84% accuracy, 0.83 F1-score), while classical models offer sub-50 ms inference, making them suitable for latency-critical applications. The Tkinter-based “Safe Keyboard” prototype validates the feasibility of pre-posting moderation with minimal user disruption. By shifting from reactive to preventive content moderation, this research contributes both technical insights and a prototype implementation toward safer online communication environments.

12. REFERENCES

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